ECE 4020-5020: Digital Signal Processing

Homework 2 (100 pts.)

Due on Thursday, March 5, 2026

Problem 1 (10 pts.):

Compute and sketch the output response $y(n)$ of the system using convolution for the following pair of input $x(n)$ and the system impulse response $h(n)$. Include the range n with your answer where the response is non-zero and is otherwise zero, respectively.

$$x(n) = \begin{cases} \frac{1}{3}n, & 0 \le n \le 6 \\ 0, & \text{otherwise} \end{cases}$$

$$h(n) = \begin{cases} 1, & -2 \le n \le 2 \\ 0, & \text{otherwise} \end{cases}$$

Problem 2 (25 pts.):

a. (10 pts.) Find the zero-input response of the system due to initial conditions $y(-1) = -\frac{1}{3}$ and $y(-2) = -\frac{2}{9}$ for the following difference equation,

$$y(n) = -6y(n-1) - 9y(n-2) + 2x(n) + 6x(n-1)$$

Hint: For a repeated root λ , the homogenous solution takes the form $y_h(n) = \left(c_1 + c_2 n\right)(\lambda)^n$

b. (15 pts.) Determine the BIBO stability using the impulse response theorem. Assume all initial conditions being zero.

$$y(n) = -\frac{1}{4}y(n-2) + 2x(n) + 2x(n-1)$$

Hint: First find the impulse response $h(n)$. For complex roots λ, you will also have complex coefficients c's in the homogenous solution $y_h(n) = c\left(\lambda\right)^n + c^*\left(\lambda^*\right)^n$, where * is conjugate. You can use the Euler identity to obtain cos and sine expressions (not required though).

Problem 3 (25 pts.):

a. (10 pts.) Determine the convolution response $y(n) = x_1(n) * x_2(n)$ using z-transforms for

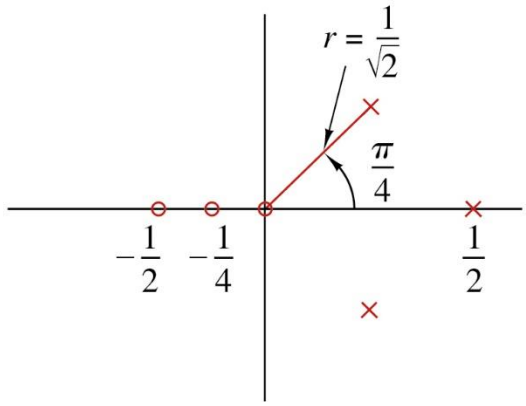
$$x_1(n) = \begin{cases} \left(\frac{1}{3}\right)^n, & n \geq 0 \\ \left(\frac{1}{2}\right)^{-n}, & n < 0 \end{cases}$$

$$x_2(n) = \left(\frac{1}{2}\right)^n u(n)$$

Hint: $x_1(n)$ is a two-sided signal so ensure you assess its correct ROC range. Accordingly, the final response may also exist for $n < 0$.

b. (15 pts.) Determine the causal signal $x(n)$ whose pole-zero plot is as follows.

Note: The constant gain of the signal is $G = \frac{1}{4}$. Hint: Use partial fractions to obtain exponential sine/cosine terms in the inverse z-transform.



Problem 4 (20 pts.):

Use z-transforms to find the output response assuming zero initial conditions (i.e., the zero-state response) for the following systems and input pairs

a. $y(n) = -0.1y(n-1) + 0.2y(n-2) + x(n) + x(n-1)$ for input $x(n) = \left(\frac{1}{3}\right)^n u(n)$

b. $y(n) = -y(n-2) + 10x(n)$ for input $x(n) = 10\cos\left(\frac{\pi}{2}n\right)u(n)$

Problem 5 (20 pts.):

For each of the following transfer functions, obtain the impulse response $h(n)$ and determine system stability i) if $h(n)$ is causal, ii) $h(n)$ is anticausal, and iii) $h(n)$ is noncausal (two-sided)

a. $H(z) = \dfrac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$

b. $H(z) = \dfrac{1}{1 - z^{-1} + \frac{1}{4}z^{-2}}$ (where applicable)

ECE 5020 Homework 2

Blaine Swieder

5 March 2026

1 Question 1

Given

$$x(n) = \begin{cases} \frac{n}{3}, & 0 \le n \le 6, \\ 0, & \text{otherwise}, \end{cases} \qquad h(n) = \begin{cases} 1, & -2 \le n \le 2, \\ 0, & \text{otherwise}. \end{cases}$$

Find and sketch $y(n) = x(n) * h(n)$ and state where it is nonzero.

1.1 Convolution Setup

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k).$$

Since $x(k) \neq 0$ only for $0 \le k \le 6$ and $h(n-k) \neq 0$ only when $-2 \le n-k \le 2$, the overlap is

$$\max(0, n-2) \le k \le \min(6, n+2).$$

Hence

$$y(n) = \sum_{k=\max(0,n-2)}^{\min(6,n+2)} \frac{k}{3}.$$

The support is from $0 + (-2) = -2$ to $6 + 2 = 8$, but $y(-2) = x(0) = 0$. Therefore

$$\boxed{y(n) \neq 0 \text{ for } -1 \le n \le 8, \qquad y(n) = 0 \text{ otherwise.}}$$

Evaluate by overlap range:

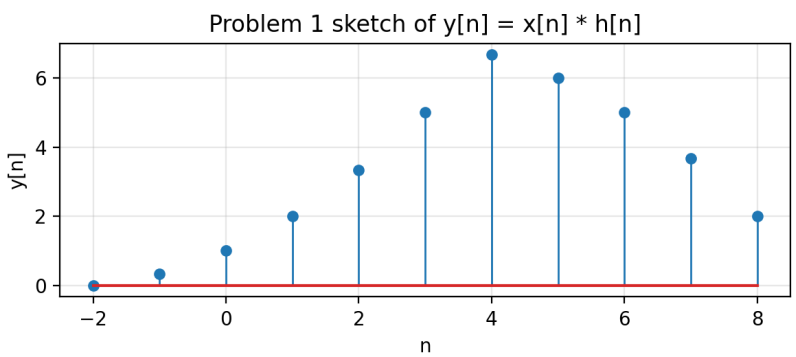
$$y(n) = \begin{cases} \frac{1}{3} \sum_{k=0}^{n+2} k, & -1 \le n \le 2, \\ \frac{1}{3} \sum_{k=n-2}^{n+2} k, & 3 \le n \le 4, \\ \frac{1}{3} \sum_{k=n-2}^{6} k, & 5 \le n \le 8. \end{cases}$$

Using $\sum_{k=0}^{m} k = \dfrac{m(m+1)}{2}$,

$$\boxed{y(n) = \begin{cases} \dfrac{(n+2)(n+3)}{6}, & -1 \le n \le 2, \\ \dfrac{5n}{3}, & 3 \le n \le 4, \\ 7 - \dfrac{(n-3)(n-2)}{6}, & 5 \le n \le 8, \\ 0, & \text{otherwise.} \end{cases}}$$

Sample values:

$$y(-1) = \frac{1}{3}, \quad y(0) = 1, \quad y(1) = 2, \quad y(2) = \frac{10}{3}, \quad y(3) = 5,$$
$$y(4) = \frac{20}{3}, \quad y(5) = 6, \quad y(6) = 5, \quad y(7) = \frac{11}{3}, \quad y(8) = 2.$$



2 Question 2

2.1 Part A

For the zero-input response, set $x(n) = 0$ in:

$$y(n) = -6y(n-1) - 9y(n-2) + 2x(n) + 6x(n-1)$$

So, it follows that:

$$y(n) + 6y(n-1) + 9y(n-2) = 0$$

From here, we can state the characteristic equation:

$$r^2 + 6r + 9 = 0 = (r+3)^2$$

So,

$$y_{zi}(n) = (c_1 + c_2 n)(-3)^n.$$

We can use $y(-1) = -\frac{1}{3}$ and $y(-2) = -\frac{2}{9}$:

$$(c_1 - c_2)(-3)^{-1} = -\frac{1}{3} \Rightarrow c_1 - c_2 = 1,$$

$$(c_1 - 2c_2)(-3)^{-2} = -\frac{2}{9} \Rightarrow c_1 - 2c_2 = -2.$$

Hence, $c_2 = 3$ and $c_1 = 4$. Therefore,

$$y_{zi}(n) = (4 + 3n)(-3)^n$$

2.2 Part B

We are given:

$$y(n) = -\frac{1}{4}y(n-2) + 2x(n) + 2x(n-1)$$

with zero initial conditions. By taking the z-transform, we get:

$$Y(z) = -\frac{1}{4}z^{-2}Y(z) + 2X(z) + 2z^{-1}X(z),$$

Then:

$$H(z) = \frac{Y(z)}{X(z)} = \frac{2(1 + z^{-1})}{1 + \frac{1}{4}z^{-2}}$$

If we factor the denominator:

$$1 + \frac{1}{4}z^{-2} = \left(1 - \frac{j}{2}z^{-1}\right)\left(1 + \frac{j}{2}z^{-1}\right)$$

Thus, the poles are at $\pm\frac{j}{2}$, whose magnitudes are $\frac{1}{2} < 1$. For the causal impulse response,

$$h(n) = c\left(\frac{j}{2}\right)^n u(n) + c^*\left(-\frac{j}{2}\right)^n u(n),$$

so

$$|h(n)| \leq C\left(\frac{1}{2}\right)^n$$

for some constant C. Hence,

$$\sum_{n=-\infty}^{\infty} |h(n)| = \sum_{n=0}^{\infty} |h(n)| < \infty$$

Therefore, by the impulse response theorem, it is the case that the system is BIBO stable.

3 Question 3

3.1 Part A

$$x_1(n) = \begin{cases} \left(\frac{1}{3}\right)^n, & n \geq 0, \\ \left(\frac{1}{2}\right)^{-n}, & n < 0, \end{cases} \qquad x_2(n) = \left(\frac{1}{2}\right)^n u(n).$$

First,

$$X_1(z) = \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^{-n} + \sum_{n=-\infty}^{-1} \left(\frac{1}{2}\right)^{-n} z^{-n} = \frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-2z^{-1}}.$$

Hence

$$X_1: \quad \frac{1}{3} < |z| < 2.$$

Also,

$$X_2(z) = \frac{1}{1-\frac{1}{2}z^{-1}}, \qquad X_2: \ |z| > \frac{1}{2}.$$

Therefore

$$Y(z) = X_1(z)X_2(z) = \left(\frac{1}{1-\frac{1}{3}z^{-1}} - \frac{1}{1-2z^{-1}}\right)\frac{1}{1-\frac{1}{2}z^{-1}},$$

with

$$Y: \quad \frac{1}{2} < |z| < 2.$$

Partial fraction decomposition gives:

$$Y(z) = -\frac{2}{1-\frac{1}{3}z^{-1}} + \frac{10/3}{1-\frac{1}{2}z^{-1}} - \frac{4/3}{1-2z^{-1}}.$$

We use the ROC to assign sidedness:

$$\frac{1}{1-az^{-1}} \overset{|z|>|a|}{\longleftrightarrow} a^n u(n),$$

$$\frac{1}{1-az^{-1}} \overset{|z|<|a|}{\longleftrightarrow} -a^n u(-n-1).$$

Thus

$$y(n) = \frac{4}{3}2^n u(-n-1) + \left[-2\left(\frac{1}{3}\right)^n + \frac{10}{3}\left(\frac{1}{2}\right)^n\right] u(n).$$

The equivalent piecewise form is:

$$y(n) = \begin{cases} \frac{4}{3}2^n, & n < 0, \\ -2\left(\frac{1}{3}\right)^n + \frac{10}{3}\left(\frac{1}{2}\right)^n, & n \geq 0. \end{cases}$$

3.2 Part B

From the pole-zero plot:

$$\text{zeros at } 0,\ -\frac{1}{4},\ -\frac{1}{2}, \qquad \text{poles at } \frac{1}{2},\ \frac{1+j}{2},\ \frac{1-j}{2}, \qquad G = \frac{1}{4}.$$

Hence

$$X(z) = \frac{\left(1+\frac{1}{4}z^{-1}\right)\left(1+\frac{1}{2}z^{-1}\right)}{4\left(1-\frac{1}{2}z^{-1}\right)\left(1-\frac{1+j}{2}z^{-1}\right)\left(1-\frac{1-j}{2}z^{-1}\right)}.$$

Since $x(n)$ is causal,

$$X:\ |z| > \frac{1}{\sqrt{2}}.$$

Partial fractions give

$$X(z) = \frac{3/4}{1-\frac{1}{2}z^{-1}} + \frac{-\frac{1}{4}-\frac{7}{16}j}{1-\frac{1+j}{2}z^{-1}} + \frac{-\frac{1}{4}+\frac{7}{16}j}{1-\frac{1-j}{2}z^{-1}}.$$

Inverse z-transform:

$$x(n) = \left[\frac{3}{4}\left(\frac{1}{2}\right)^n + \left(\frac{1}{\sqrt{2}}\right)^n\left(-\frac{1}{2}\cos\frac{\pi n}{4} + \frac{7}{8}\sin\frac{\pi n}{4}\right)\right]u(n).$$

4 Question 4

4.1 Part A

Given

$$y(n) = -0.1y(n-1) + 0.2y(n-2) + x(n) + x(n-1), \qquad x(n) = \left(\frac{1}{3}\right)^n u(n).$$

With zero initial conditions,

$$Y(z)\left(1+0.1z^{-1}-0.2z^{-2}\right) = (1+z^{-1})X(z), \qquad X(z) = \frac{1}{1-\frac{1}{3}z^{-1}}.$$

By factoring, we obtain:

$$1+0.1z^{-1}-0.2z^{-2} = (1-0.4z^{-1})(1+0.5z^{-1}).$$

So

$$Y(z) = \frac{1+z^{-1}}{(1-0.4z^{-1})(1+0.5z^{-1})\left(1-\frac{1}{3}z^{-1}\right)}.$$

Through partial fraction decomposition, we get:

$$Y(z) = \frac{28/3}{1-0.4z^{-1}} - \frac{1/3}{1+0.5z^{-1}} - \frac{8}{1-\frac{1}{3}z^{-1}}.$$

Therefore,

$$y(n) = \left[\frac{28}{3}(0.4)^n - \frac{1}{3}(-0.5)^n - 8\left(\frac{1}{3}\right)^n\right]u(n).$$

4.2 Part B

Given:

$$y(n) = -y(n-2) + 10x(n), \qquad x(n) = 10\cos\left(\frac{\pi n}{2}\right)u(n)$$

If we take the z-transform, it follows that:

$$Y(z)(1+z^{-2}) = 10X(z).$$

Since

$$\mathcal{Z}\left\{\cos\left(\frac{\pi n}{2}\right)u(n)\right\} = \frac{1}{1+z^{-2}},$$

we have

$$X(z) = \frac{10}{1+z^{-2}}, \qquad Y(z) = \frac{100}{(1+z^{-2})^2}.$$

By using the standard inverse transform, it follows that:

$$y(n) = 100\left(\frac{n}{2}+1\right)\cos\left(\frac{\pi n}{2}\right)u(n).$$

5 Question 5

5.1 Part A

Given:

$$H(z) = \frac{1}{1-1.5z^{-1}+0.5z^{-2}} = \frac{1}{(1-z^{-1})(1-0.5z^{-1})}$$

By using partial fraction decomposition, we get:

$$H(z) = \frac{2}{1-z^{-1}} - \frac{1}{1-0.5z^{-1}}$$

Here are my answers:

1. **Causal**: Note $|z| > 1$. So,

$$\boxed{h(n) = \left[2 - \left(\frac{1}{2}\right)^n\right]u(n).}$$

   Since

$$\sum_{n=0}^{\infty} 2 = \infty,$$

   the given system is **unstable**.

2. **Anticausal**: Note $|z| < 0.5$. So,

$$\boxed{h(n) = \left[-2 + \left(\frac{1}{2}\right)^n\right]u(-n-1).}$$

   This is not absolutely summable, so the system is **unstable**.

3. **Noncausal**: Note $0.5 < |z| < 1$. So,

$$\boxed{h(n) = -2u(-n-1) - \left(\frac{1}{2}\right)^n u(n).}$$

Again, this is not absolutely summable, so the system is **unstable**.

5.2 Part B

Given:

$$H(z) = \frac{1}{1 - z^{-1} + \frac{1}{4}z^{-2}} = \frac{1}{\left(1 - \frac{1}{2}z^{-1}\right)^2}$$

Here are my answers:

1. **Causal**: Note $|z| > 0.5$, it follows that:

$$\boxed{h(n) = (n+1)\left(\frac{1}{2}\right)^n u(n)}.$$

Since

$$\sum_{n=0}^{\infty}(n+1)\left(\frac{1}{2}\right)^n < \infty,$$

the system is **stable**.

2. **Anticausal**: Note $|z| < 0.5$. A valid left-sided form is:

$$\boxed{h(n) = -(n+1)\left(\frac{1}{2}\right)^n u(-n-2)}.$$

This is not *absolutely summable*, so the system is **unstable**.

3. **Noncausal**: There is no annular ROC since both poles have the same magnitude 0.5, so this case is **not applicable**.